

Tianxin Hu

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Educational Background

Nanyang Technological University (Master of Science) - School: Electrical and Electronic Engineering - Major: Computer Control & Automation - Dissertation: Minimizing Swept Area for AGVs in Singapore Seaport - GPA: 95.00/100	Aug 2024 – Jan 2026
Wuhan University of Science and Technology (Bachelor of Engineering) - School: Automobile and Traffic Engineering - Major: Vehicle Engineering (Industrial Planning) - GPA: 88.30/100 (Class Ranking: 3/32 Department Ranking: 8/157)	Sept 2020 – Jun 2024

Research & Academic Experience

Chery Robotics — Humanoid Robot Locomotion Development Supervisor: Professor Lihua Xie, Dr. Shenghai Yuan Position: Research Assistant Project: Industry-oriented humanoid robotics project targeting language-guided embodied task execution, such as object fetching in real-world environments Responsibilities: Responsible for the locomotion module of Chery's humanoid robot, with a focus on reinforcement learning-based humanoid locomotion training and deployment	Mar 2026 – Present
AIBOT Technology Pte. Ltd. — Autonomous Navigation System Development Supervisor: Professor Lihua Xie, Dr. Shenghai Yuan Position: Navigation Module Lead Responsibilities: Developed the full-stack navigation module for the company's autonomous mobile robot, responsible for cross-floor navigation, elevator interface integration, and dynamic obstacle avoidance within residential and commercial environments ✓ Achievements: Achieved fully autonomous navigation and delivery within multi-story residential complexes	Jul 2025 – Present
Research Student — NTU Internet of Things Lab & Center for Advanced Robotics Technology and Innovation (CARTIN) Supervisors: Professor Lihua Xie, Dr. Shenghai Yuan Research Focus: Developed optimization-based motion planning methods for multi-axle autonomous vehicles to minimize swept volume during turning maneuvers and reduce tire wear under dynamic constraints ✓ Achievements: Published five papers on vehicle planning and control (three under review) [J1-J3]	Aug 2024 – Mar 2026
Xin Chi Technology — Research and Development of Autonomous Driving System for Multi-Axle Heavy-Duty Transport Vehicles Supervisor: Senior Engineer Xiangrong You Position: Technical Director & Corporate Legal Representative Responsibilities: Led R&D of autonomous driving systems for heavy-duty vehicles, covering mathematical modeling, ROS architecture, microcontroller development, and both simulation and prototype validation ✓ Achievements: Contributed to multiple patents [P1-P4] and received several national awards [A1-A5]	Jun 2022 – Jun 2024
Research and Development of Autonomous Driving System for Formula Racing Supervisors: Associate Professor Chao Deng, Dr. Junyi Zou Position: Leader of the Unmanned Group Responsibilities: Led the development of the autonomous driving system for Formula Student Autonomous China (FSAC), covering visual detection, mapping, planning, and tracking ✓ Achievements: Received the FSAC Third Prize and Best Rookie Award [A6], and published a paper on it [J4]	Feb 2021 – Jun 2024

Publications

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- [J1] **Swept Volume-Aware Trajectory Planning and MPC Tracking for Multi-Axle Swerve-Drive AMRs** May 2025
IEEE International Conference on Robotics and Automation (ICRA 2025)
Tianxin Hu, Shenghai Yuan, Ruofei Bai, Xinhang Xu, Yuwen Liao, Fen Liu, Lihua Xie
DOI: 10.1109/ICRA55743.2025.11127914
- [J2] **Tire Wear-Aware Trajectory Tracking Control for Multi-Axle Swerve-Drive Autonomous Mobile Robots** Jun 2025
Journal of Automation and Intelligence (JAI)
Tianxin Hu, Xinhang Xu, Thien-Minh Nguyen, Fen Liu, Shenghai Yuan, Lihua Xie
DOI: 10.1016/j.jai.2025.05.003
- [J3] **TGSFormer: Scalable Temporal Gaussian Splatting for Embodied Semantic Scene Completion** Mar 2026
IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2026)
Rui Qian, Haozhi Cao, Tianchen Deng, Tianxin Hu, Weixing Guo, Shenghai Yuan, Lihua Xie
DOI: 10.48550/arXiv.2512.00300
- [J4] **Improvement of YOLOv5 Algorithm for Formula Racing Based on TensorRT under ROS Platform** May 2023
Agricultural Equipment & Vehicle Engineering
Tianxin Hu, Chao Deng, Junjie Ma, Wang Liu
DOI: 10.3969/j.issn.1673-3142.2023.05.004

Patents

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- [P1] **(Granted) Multi-axle all-wheel steering vehicle vector-division correction method and system** Sep 2025
Tianxian Hu, Xiangrong You, Dongmei Wang, Xinhe Long, Xinyu Wang, Shuang Liu, Yutong Fu, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Patent No.: CN116534005B
- [P2] **(Granted) A system and method for automatically correcting the deviation of a beam transport vehicle in a tunnel** Jul 2023
Xiangrong You, Enze Jin, Dongmei Wang, Yuran Huang, Tianxian Hu, Shuai Zhu, Beijia Zhang
Patent No.: CN114633801B
- [P3] **(Pending) Energy-saving system and method for beam transport vehicle based on traveling pressure control** Feb 2024
Wei Li, Xiangrong You, Dongmei Wang, Xinhe Wang, Peng Cheng, Yuming Liu, Shuang Hu, Tianxian Hu, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Application No.: CN117508238A
- [P4] **(Pending) Method, system and equipment for planning obstacle avoidance path of multi-shaft all-wheel steering beam transporting vehicle** Dec 2023
Xiangrong You, Yucheng Yin, Liwei Wang, Dongmei Wang, Tianxian Hu, Shuang Liu, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Application No.: CN117168481A

Awards

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- [A1] **(International Level) Bronze Award of the 9th China International College Students Innovation Competition** Apr 2024
Organizing Committee of China College Students "Internet+" Competition
- [A2] **(National Level) The First Prize of the 18th "Challenge Cup" National Undergraduate Curricular Academic Science and Technology Works** Oct 2023
China Association for Science and Technology, Ministry of Education of the PRC, Chinese Academy of Social Sciences
- [A3] **(International Level) Bronze Award of the 8th China College Students "Internet+" Innovation and Entrepreneurship Competition** Apr 2023
Organizing Committee of China College Students "Internet+" Competition
- [A4] **(National Level) Second Prize of the 15th China College Student Computer Design Competition** Jul 2022

Organizing Committee of Chinese College Student Computer Design Competition

[A5] (National Level) Second Prize of the 15th Energy Saving and Emission Reduction Competition Aug 2022

China College Student Energy Conservation and Emission Reduction Social Practice and Technology Competition Committee

[A6] (National Level) Third Prize and Best Rookie Award of Formula Student Competition Oct 2022

China Society of Automotive Engineers, Organizing Committee of FSAC

Technical Expertise

Programming: C, C++, Python, XML (URDF/Launch)

Robotics & Simulation: ROS/ROS2, Isaac Sim/Lab, Gazebo Classic & GZ Sim, RViz, URDF/SDF modeling

Embedded Systems: NVIDIA Jetson Series, STM32, ESP8266, 8051; FreeRTOS, LVGL, U8G2

Hardware Design: Altium Designer, Proteus; multilayer PCB design and soldering

Modeling, Simulation & CAD: MATLAB, COMSOL, SolidWorks, CATIA, Blender, AutoCAD